

Networking Device & Inter Networking Devices:

#1: Hub:



Hub



Types of Hubs:

- Active Hub
- Passive Hub

Use of Hub:

- Multiple devices can be connected by the means of a hub.
- Simple to use.
- Simply broadcast all incoming data packets on active ports.
- Layer 1 device.
- Half Duplex device
- Slow
- Divides the bandwidth if network connection increases.

#2: Switches:

There are few popular companies that design the Switches like Juniper, Nortel, DLink, Cisco, NETGEAR, NVIDIA, Huawei, LinkSys.

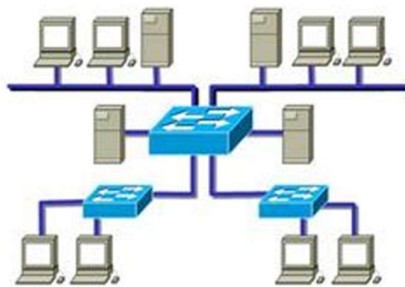


- ✓ Switches are devices that are operating a secondary data link layer in the OSI model.
- ✓ L2 Device.
- ✓ Maintains MAC Address table.
- ✓ First time do frame broadcast, then unicast and/or multicast depending upon the need.
- ✓ Full Duplex device.
- ✓ Faster than Hub.
- ✓ Does not divides the bandwidth if network connection increases.

Types of switches:

- **Managed switch:** can be configured, costlier than unmanaged switch.
- **Unmanaged switch:** cannot be configured, simply provide basic networking

Three Switch Functions



- Address learning
- Forward/filter decision
- Loop avoidance



Comparison between Hub Switch & Router

template	Hub	Switch	Router
Layer	Physical layer	Data link layer	Network layer
Function	To connect a network of personal computers together, they can be joined through a central hub	Allow connections to multiple devices, manage ports, manage VLAN security settings	Direct data in a network

Data Transmission form	electrical signal or bits	frames	packets
Port	4/8/16 ports	multi-port, usually between 4 and 256	2/4/5/8 ports
Transmission type	Data flooding, broadcast	First broadcast, then unicast and/or multicast depends on the need	The router doesn't forward broadcast. But it can generate a broadcast message
Device type	Non-intelligent device	Intelligent device	Intelligent device
Used in(LAN, MAN, WAN)	LAN	LAN	LAN, MAN, WAN
Transmission mode	Half duplex	Full duplex	Full duplex

#3: Router:

There are few popular companies that design the routers --- **Cisco, 3Com, HP, Juniper, D-Link, Nortel, Huawei** etc



- ✓ Router is defined as a networking device that sends data packets further between computer networks.
- ✓ They are able to perform traffic directing functions over the internet.
- ✓ Router never allow broadcast beyond the network boundary.
- ✓ Maintains a routing table.
- ✓ Comes with its IOS(Internetworking Operation System)

Functions of Router:

- Internetwork communication
- Path Selection
- Packet Switching
- Packet Filtering

Series of Routers:

- **Cisco ISR:** (integrated Service Routers): typically used as **internet edge** for customers with small/medium networks.
- **Cisco ASR:** (Aggregation Services Routers): typically used as **internet edge** for enterprise networks and service providers.

Types of routers:

Wired and wireless routers

- ✓ These routers are mostly used in homes and small offices.
- ✓ Wired routers share data over cables and create wired local area networks (LANs), while wireless routers use antennas to share data and create wireless local area networks (WLANs).

Edge routers

- ✓ This is a wired or wireless router that distributes data packets between one or more networks but not within a network.
- ✓ As their name indicates, edge routers are placed at the edge or boundary of networks, and typically connect to Internet service providers (ISPs) or other organizations' networks.
- ✓ Their job is to keep your network communicating smoothly with other networks.

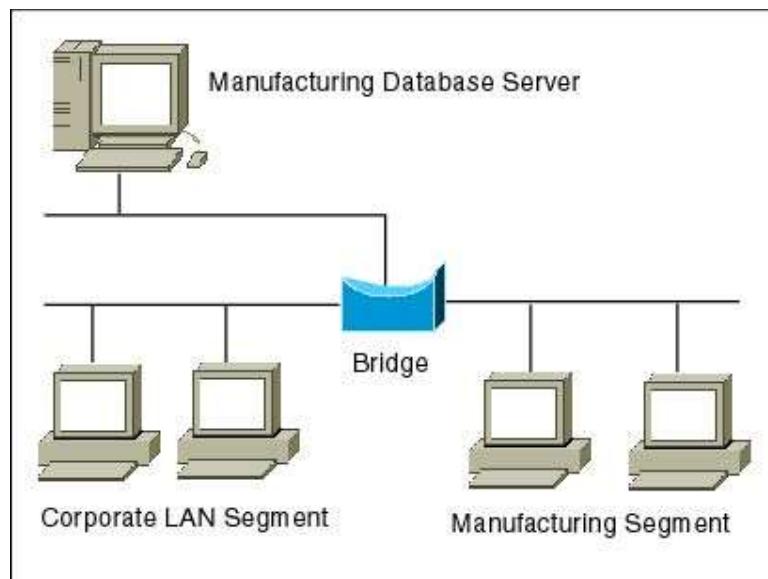
Core routers

- ✓ A **core router** is a router designed to operate in the Internet backbone, or core.
- ✓ To fulfil this role, a router must be able to support multiple telecommunications interfaces of the highest speed in use in the core Internet and must be able to forward IP packets at full speed on all of them.
- ✓ It must also support the routing protocols being used in the core

Virtual routers

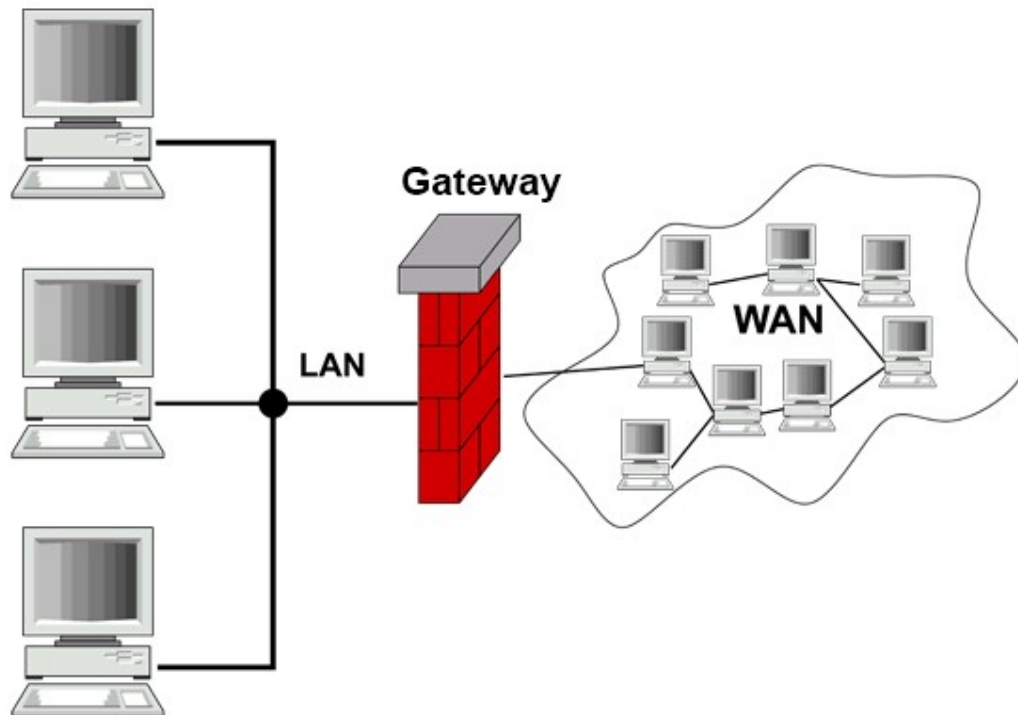
- ✓ Unlike physical routers, virtual routers are **pieces of software** that allow computers and servers to operate like routers.
- ✓ They'll share data packets just as physical routers do.

#4: Bridge:



- ✓ A bridge operates at the data link layer.
- ✓ A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination.
- ✓ It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

#5: Gateway:

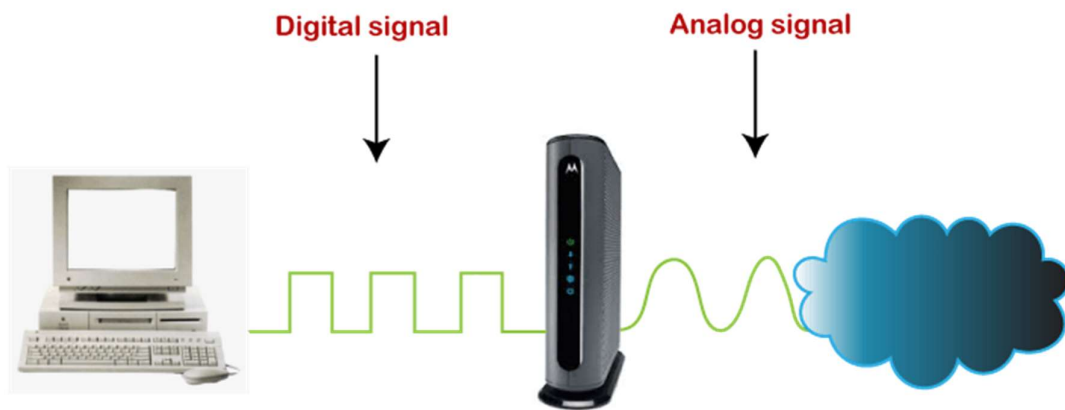


- ✓ A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models.
- ✓ They work as messenger agents that take data from one system, interpret it, and transfer it to another system.
- ✓ Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

#6: Modem:

A modem is defined as a network device that both modulates and also demodulates analog carrier signals known as signals or sine waves, for encoding and decoding digital information for processing. Modems efficiently accomplish both

tasks.



#7: Repeater:



- ✓ A repeater is implemented in a system's network to facilitate
- ✓ A repeater is implemented in computer networks to expand the coverage area of the network, re propagate a weak or broken signal and or service remote nodes.
- ✓ Repeaters amplify the received/input signal to a higher frequency domain so that it is reusable, scalable and available.

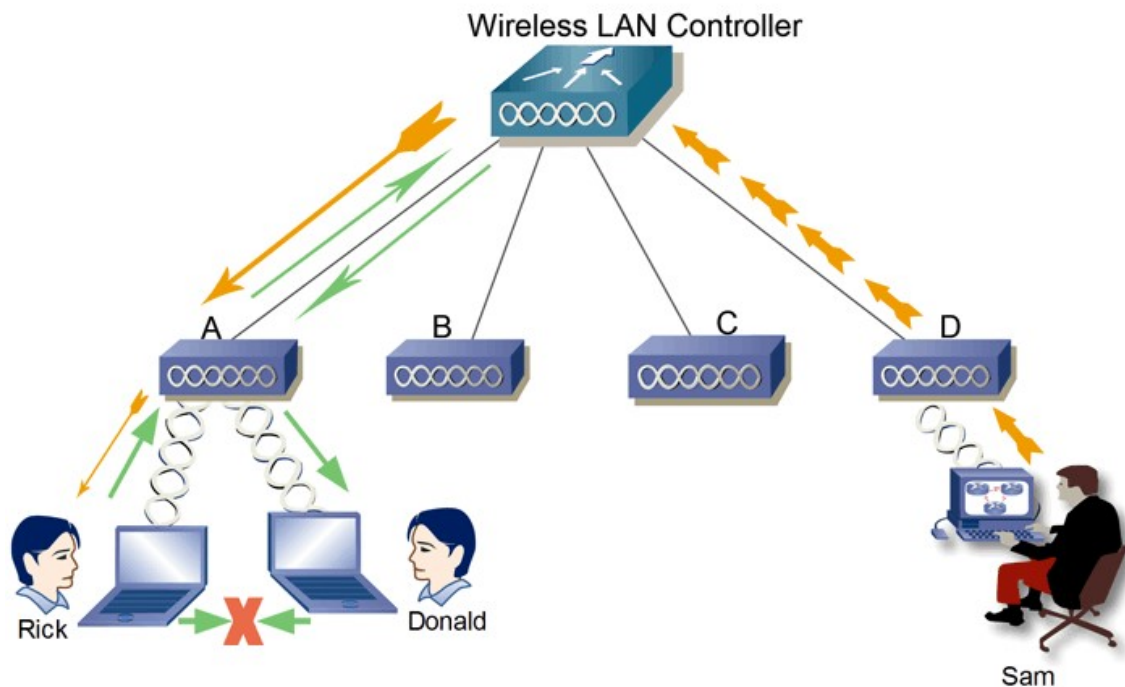
#8: Access Point:

- ✓ An access point is identified as a device that creates a wireless local area network, or WLAN, that often exists in an office or large building
- ✓ An access point establishes a connection with a wired router, switch, or hub via an Ethernet cable, and projects a Wi-Fi signal to a designated area.



#9: WLC - Wireless LAN Controller:

Cisco Wireless LAN Controller (WLAN), wireless controller, provides wireless performance of all mobile devices, offers limited hotspot coverage and more.



Need for WLC:

WLCs (Cisco WLC) are commonly used to have control over one's routers, switches, firewalls, gateways etc.

Monitoring Cisco WLC helps you determine the performance and efficiency of every device.

#10: NIC - Network Interface Card:



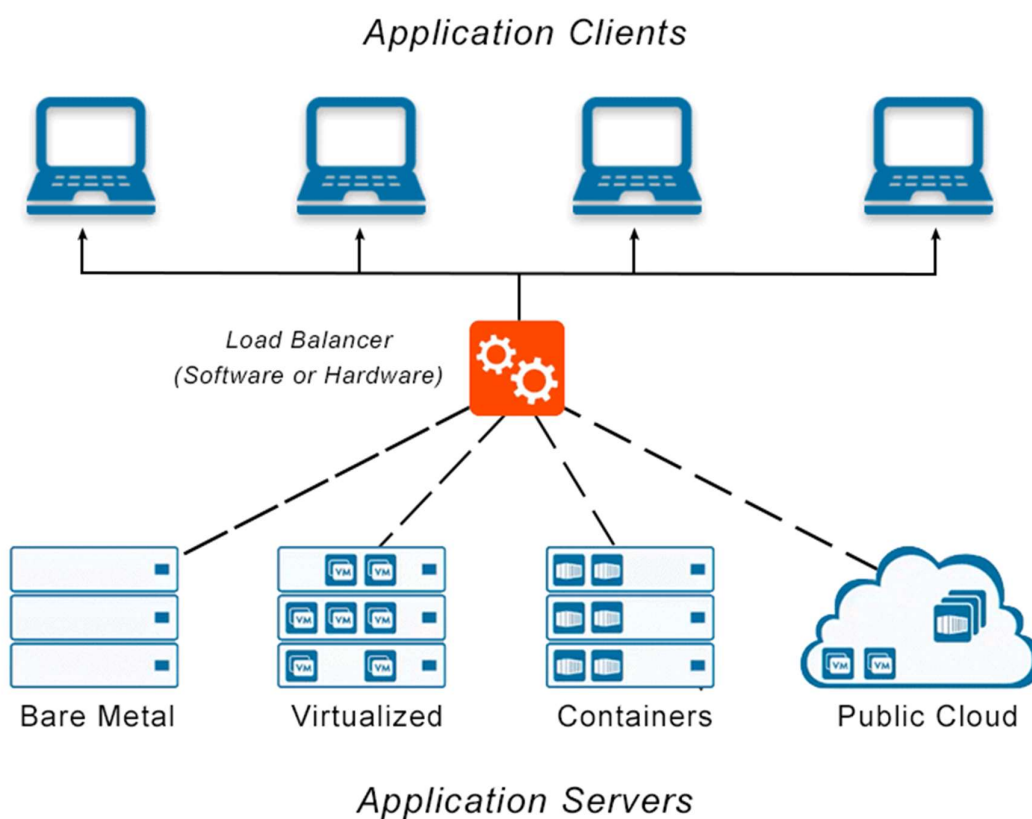
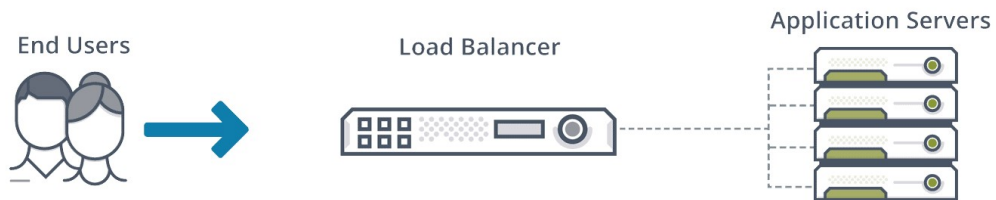
A network interface card (NIC) is a hardware component, typically a circuit board or chip, which is installed on a computer so it can connect to a network.

Need of NIC:

- ✓ A NIC is competent enough to provide computers with a dedicated, full-time connection from the network.

#11: Load Balancer:

- ✓ Load balancer is defined as the **traffic cop**, sitting in front of any server and routing client requests across all servers that are capable of fulfilling the former request.
- ✓ It is done in a way that maximizes speed and capacity utilization and ensures that no server happens to be overburdened.
- ✓ Load balancers let you evenly distribute network traffic to prevent failure that can be caused by overloading a particular resource.



TYPES OF NETWORK DEVICES



HUB

A hub joins multiple devices on the same LAN, broadcasting messages to all ports without examining frames.



SWITCH

A network switch forwards data to its proper destination, examining a packet's MAC address info to determine the intended device.



ROUTER

A router directs data requests from one network to another, using a packet's IP address to forward it to its destination.



BRIDGE

A network bridge acts as an interconnection between two LANs, creating a single network from separate LANS.



GATEWAY

A gateway connects discrete networks and translates packet data so it can travel between the systems.



MODEM

A modem modulates and demodulates signals between devices, such as analog to digital.



REPEATER

A repeater strengthens a signal and retransmits it along to its destination.



ACCESS POINT

An AP is a device that sends and receives data wirelessly over radio frequencies.